

Practice Exam: Written Answers

Diagnostic Technician practice questions · Set A (1 to 20) and Set B (21 to 40)

These are sample answers to all 40 questions, written the way a well-prepared candidate would write them on paper: in plain words, with numbered steps in the order you would really do them, the commands named, and the reason for each step.

They are fuller than what you can write in about 3 minutes per question, so treat them as the target, not a script. On the day, the numbered steps, the right terms and commands, and ending with verify and document are what earn the credit. Take each set yourself first, with a 60-minute timer and no notes, then compare.

What graders look for: the right terms and commands, evidence before changing anything, simple physical checks early, a logical order, isolating by changing one thing at a time, verifying the fix, and documenting the result.

Set A

Questions 1 to 20

1

In your own words, what is Linux? Name two Linux distributions used on servers, and give four specific reasons Linux runs most data center servers.

Linux is an operating system. Strictly speaking, "Linux" is the kernel, the core part that talks to the hardware, and it gets packaged with tools into distributions. It is open source, so the code is public and free to use.

Two server distributions: Red Hat Enterprise Linux (RHEL) and Ubuntu Server. Rocky Linux and SUSE are others.

Why it runs most data center servers:

1. It is stable. A server can run for months without a reboot.
2. It is low cost. Most distributions are free, and the paid ones like RHEL charge for support, not for the software.
3. It is built for remote work. Everything is done from the command line and scripts over the network, with no screen needed, so one person can manage hundreds of servers.
4. It is open about the hardware. Tools like `lspci` and `dmesg` show exactly what the hardware is doing, which is what we need for diagnostics.

It is also secure and efficient with frequent updates, and most AI and cloud software is built for Linux first.

2

Here is part of the `ip a` output from a server. List every piece of information you can read from it. Then name two things this command does not show, and where you would find them.

```
2: ens1f0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc mq state UP  
    link/ether 3c:ec:ef:12:34:56 brd ff:ff:ff:ff:ff:ff  
    inet 10.20.5.14/22 brd 10.20.7.255 scope global dynamic ens1f0
```

What I can read from it:

- It is interface number 2, called `ens1f0` (Ethernet, slot 1, port 0). The name can change if the card moves slots.
- `UP` means the interface is enabled. `LOWER_UP` means it has a physical link, so the cable and switch port are working. `BROADCAST` and `MULTICAST` mean it can send those kinds of traffic.
- `mtu 9000` means jumbo frames are on. The switch port has to match, or large packets will fail.
- `qdisc mq` is the queueing setup (multi-queue), and `state UP` confirms the link is up.
- `link/ether 3c:ec:ef:12:34:56` is the MAC address. `ff:ff:ff:ff:ff:ff` is the broadcast MAC.
- `inet 10.20.5.14/22` is the IPv4 address with a /22 mask (255.255.252.0), so the local subnet runs from 10.20.4.0 to 10.20.7.255.
- `brd 10.20.7.255` is the broadcast address for that subnet.
- `scope global` means it is a normal, routable address.
- `dynamic` means the address came from DHCP, not set by hand. A static address would say "forever" for the lease.

What it does not show:

- The default gateway. I would use `ip route`.
- The DNS servers. They are in `/etc/resolv.conf`.
- It also does not show link speed, which `ethtool ens1f0` shows.

3

A coworker says: "Serial Over LAN is just a slower version of SSH." Explain why that is wrong. Describe how each one reaches the server, what each depends on, and give one situation where only SOL will help.

That is wrong because they reach the server in completely different ways and show you different things.

SSH is in-band. It runs inside the operating system over the server's normal network and talks to the `sshd` service. It needs the OS to be booted, the network to be working, and `sshd` to be running. You get a login shell once the OS is up.

SOL is out-of-band. It goes through the BMC over the management network and gives you the server's serial console. The BMC runs on standby power and does not depend on the OS at all. Through SOL you can watch BIOS/POST (if console redirection is on), the GRUB menu, kernel boot messages and kernel panics, none of which SSH can ever show.

So SSH talks to the operating system, and SOL talks to the server's console through the BMC. If the OS or its network is broken, SSH is gone but SOL still works.

Situation where only SOL helps: a server drops into emergency mode at boot because of a bad entry in `/etc/fstab`. The network never comes up and `sshd` never starts, so SSH fails. With `ipmitool sol activate` I can watch the boot, log in at the emergency prompt and fix the `fstab` line.

4

A brand-new server is racked and plugged in. Describe every DHCP message exchanged before it has an address, who sends each one, which are broadcasts, what settings the server ends up with, and what you would see if no DHCP server answered.

There are four messages, known as DORA:

1. Discover: the new server broadcasts "I need an address." It has no IP yet, so it sends from 0.0.0.0 to the broadcast address 255.255.255.255.
2. Offer: a DHCP server replies with an offer of an IP address and settings.
3. Request: the server broadcasts that it wants to use that offer. Because it is a broadcast, any other DHCP server that made an offer knows it was not taken.
4. Acknowledge: the DHCP server confirms the lease, and the server configures itself.

Discover and Request come from the client and are broadcasts. Offer and Acknowledge come from the DHCP server (sent as broadcast or unicast depending on the setup). Broadcasts do not cross routers, so if the DHCP server is on another subnet, the router needs a DHCP relay.

The server ends up with an IP address, a subnet mask, a default gateway, DNS servers and a lease time. The lease is renewed partway through.

If no DHCP server answers, the server either has no IPv4 address at all or gives itself a self-assigned 169.254.x.x address. In `ip a` there would be no `inet` line or a 169.254 address with no "dynamic", and the server could not reach anything beyond its own link.

5

What determines the bandwidth of a PCIe link? List the factors. Then estimate the bandwidth in each direction of a Gen 4 x8 link and show your math.

Factors:

1. The generation, which sets the transfer rate per lane. Gen 3 is 8 GT/s, Gen 4 is 16 GT/s, Gen 5 is 32 GT/s. Each generation doubles the last.
2. The width, meaning the number of lanes: x1, x4, x8 or x16.
3. Encoding overhead. Gen 3 to 5 lose about 1.5% to encoding (Gen 1 and 2 lost 20%).
4. The weakest part of the path. The link runs at the best speed and width that both ends and everything in between support, so a Gen 5 card in a Gen 4 slot runs at Gen 4.

5. What the link actually trained to (LnkSta in `lspci -vv`), and how the slot is wired. An x16 slot may only be wired x8.
6. Shared links. Devices behind a PCIe switch share one upstream link to the CPU.

Math for Gen 4 x8:

16 GT/s x 8 lanes = 128 Gb/s.

128 / 8 bits per byte = 16 GB/s.

Minus about 1.5% for encoding = about 15.75 GB/s.

So roughly 16 GB/s in each direction. PCIe sends and receives at the same time, so about 32 GB/s total both ways.

6

A manager asks why servers use ECC memory when most desktops do not. Explain how ECC works, what it can and cannot fix, and why the log of corrected errors matters.

ECC means Error-Correcting Code. ECC memory stores extra check bits along with the data. When the data is read back, the memory controller uses those check bits to see if anything changed.

- If one bit has flipped, ECC detects it and fixes it on the spot. That is a correctable error. The server keeps running and logs it.
- If two bits are wrong, ECC can detect it but cannot fix it. That is an uncorrectable error, and the server will usually stop or crash on purpose rather than use bad data.
- It cannot save you from a DIMM that has failed completely.

Why servers need it: a server runs 24/7 with a lot of memory, so bit flips will happen. Without ECC a flipped bit is silent data corruption, a wrong result or a random crash, and you would never know why. A home PC crashing once in a while is annoying; a database server quietly corrupting data is a real problem.

The corrected error log matters because it is an early warning. A DIMM that logs more and more corrected errors is likely to fail. You can see it in the BMC event log (for example "Correctable ECC logging limit reached, DIMM C1"), `dmesg | grep -i edac` or `ras-mc-ctl --summary`, and replace that DIMM in a planned window before it causes an uncorrectable error and an outage.

7

Write one command that searches every log file under `/var/log`, including subfolders, for the word `nvme`, ignoring upper and lower case and showing line numbers. Explain what each option does, and how you would count the matches instead.

```
sudo grep -rin "nvme" /var/log/
```

- `-r` searches recursively, so the folder and every subfolder.
- `-i` ignores case, so it matches `nvme`, `NVMe` and `NVME`.
- `-n` shows the line number of each match. With `-r` it also shows which file the line is in.

- "nvme" is the word to find and /var/log/ is where to look. I use sudo because some logs can only be read by root.

To count the matches instead, swap -n for -c:

```
sudo grep -ric "nvme" /var/log/
```

That prints a count of matching lines for each file. For one total I would pipe it into wc: `sudo grep -ri "nvme" /var/log/ | wc -l`

8

A Linux server stopped answering ping and SSH right after it was moved to a new rack position. What is your very first check, and what is the reason for it? Then walk through the rest of your process in order.

First check: is the server actually up? I would use the BMC (virtual console or SOL) to look at the console and the power status. You cannot SSH into a server that is off or crashed, and after a move it may not have powered back on, may be stuck in POST, or may be sitting at an error. The BMC tells me whether this is a server problem or a network problem before I start pulling cables.

Then, in order:

1. Physical: the network cable is plugged in at both ends, the link lights are on, the transceiver is seated, and the cable is in the right switch port for the new position. `ethtool ens1f0` should show "Link detected: yes", and `ip a` shows LOWER_UP or NO-CARRIER. Try a known-good cable or port.
2. NIC and driver: `lspci`, `lspci -k` and `dmesg`, since a card or riser can come loose during a move.
3. IP address: `ip a`. Is there an address? A 169.254 address means DHCP failed, which can happen if the new switch port is on a different VLAN.
4. Gateway: `ip route`, ping the gateway, then ping past it and use `traceroute`.
5. Names: test by IP and by name.
6. Service and firewall: `systemctl status sshd`, `ss -tulpn`, firewall rules.
7. Ask what changed with the move (new port, VLAN, cable), test from another machine, fix it, confirm ping and SSH work, and document every check.

9

A technician installs a new 100G network card and says it is dead because `lspci` does not show it. What would you check, in order, before agreeing?

1. Make sure we are looking properly. A 100G Mellanox card shows as "Ethernet controller" (or "Infiniband controller"), so I would run `lspci | grep -i -E "ethernet|mellanox|infiniband"` and compare the count with what the build sheet expects.
2. Check the BIOS and BMC inventory. If they cannot see the card either, the problem is physical or firmware. If they see it but Linux does not, it is something else.
3. Physical, powered off with an ESD strap: card fully seated and latched, riser seated, riser cable connected, any power cable connected, no bent pins or damage, and it is in the slot the build calls for.

4. BIOS settings: the slot is enabled, bifurcation is right for that slot, and the link speed is not forced to something odd.
5. Topology: compare `lspci -tv` with the topology diagram. If the port for that slot is there with nothing under it, suspect the card, its seating or the slot. If the whole branch is missing, suspect the riser, cable, switch or retimer. Are other devices on the same riser missing too?
6. Check `dmesg` and the BMC event log for PCIe errors.
7. Swap test: put the card in a known-good slot, or a known-good card in this slot. If the problem follows the card, the card is bad.

Only then would I agree it is dead, tag it, document it and start an RMA.

10

A test fails roughly once every 15 runs, and you cannot make it fail on demand. List the data you would gather, and explain how you would turn it into a failure you can reproduce.

Data to gather:

- The exact error message and which test step fails, every time.
- Timestamps and how often (which runs failed).
- Which tests pass and which fail.
- Temperature and load at the time of each failure.
- Logs from each failure: `dmesg`, `journalctl` and the BMC SEL.
- Serial numbers, plus firmware and driver versions.
- Whether other units, or other slots, show the same thing.

Turning it into something I can reproduce:

1. Look for a pattern in that data: only under heavy load, only when it is hot, only late in a long run, only in one slot or one unit.
2. Run the test in a loop and log every run, for example `for i in $(seq 1 50); do ./test.sh >> results.txt; done`, while watching `dmesg -w` and the temperatures, so I catch the failure with its logs.
3. Push the suspected condition on purpose: a stress test, more load or more heat.
4. Look for corrected errors (PCIe AER or ECC) that show a weak link before it actually fails.
5. Check seating and cables, then change one variable at a time and rerun enough cycles to trust the result. At 1 in 15, twenty clean runs does not prove anything; I would want a hundred or more.
6. Compare with a golden unit, and document every step and result.

11

A server keeps logging correctable memory errors. Walk through how you would decide whether the DIMM, the slot, or the CPU is the cause.

1. Record the evidence first: which slot (from the BMC SEL, `ras-mc-ctl --summary` or `dmesg | grep -i edac`), the error type and count, the timestamps, and the DIMM serial number from `dmidecode -t memory`.
2. Simple checks: the DIMM is seated with both latches closed, it is the correct part, it is in the right slot for the population rules, and the BIOS is current.
3. Look for a pattern. If errors are showing up on many DIMMs on the same channel or the same CPU, that points to the CPU's memory controller, CPU seating or bent socket pins, not one DIMM.
4. If it is one slot, do a swap test. Move the suspect DIMM to a slot on a different channel, and put a known-good DIMM in the suspect slot. Clear the logs and run a memory test (vendor diagnostics, MemTest86 or stress-ng).
 - If the errors follow the DIMM to its new slot, the DIMM is bad.
 - If the errors stay in the original slot with the good DIMM in it, the problem is the slot, the board or the CPU.
5. If it is on the slot side, inspect the slot and check the CPU seating and socket pins, then try a known-good CPU if needed.
6. Change one thing at a time, stress test after each change, and document the slots, serials and results.

12

A coworker reseats a card, replaces its cable, and updates its firmware all at once, and the problem goes away. Explain what is wrong with that approach and what could go wrong later because of it.

The problem is that three things changed at once, so nobody knows which one fixed it. It could have been the reseat, the cable or the firmware.

What can go wrong later:

- If the problem comes back, on this unit or another one, we still do not know the real cause, so we do not know what to fix.
- A good part may get blamed. If the cable was fine, it gets scrapped or sent back as bad, which wastes money and gives the supplier wrong data.
- If the firmware was the real fix, every other unit on the old firmware has the same problem and nobody knows to update them. If it was seating, it may be an assembly problem that keeps happening.
- Changing several things at once can introduce a new problem and make it even harder to sort out.
- The result is not repeatable, so it cannot be written down as a known fix.

The right way is to change one thing, retest, and only move on to the next change if it still fails. Then you know the real cause, you avoid replacing good parts, and the fix can be documented and repeated.

13

A technician writes “Root cause: GPU not detected” on a repair ticket. Explain why that is a symptom and not a root cause. Then use the 5 Whys to trace an example of your own down to a process root cause.

"GPU not detected" is what was observed, so it is the symptom. The root cause is why it happened. If you only fix the symptom, for example reseal the GPU and ship the unit, the same thing will happen again on the next unit.

Example with the 5 Whys:

1. Why was the GPU not detected? Its auxiliary power cable was not connected.
2. Why was it not connected? The cable was routed in a way that could not reach the GPU's power connector.
3. Why was it routed that way? The assembler followed the work instruction.
4. Why did the work instruction show that routing? It showed an old layout.
5. Why did it show an old layout? It was not updated after a design change to the chassis.

So the root cause is the out-of-date work instruction, which is a process problem, not this one unit. The fix is to update the work instruction, make sure design changes trigger updates to the instructions, and check the other units built with the old version.

14

What should be documented after a repair, and why does documentation matter to people other than you?

What I would document:

- The unit serial number (and the ticket or traveler).
- The symptom and the exact error messages.
- The steps I took and the result of each one.
- Parts changed, with the serial numbers of the old and new parts.
- Firmware and driver versions, before and after if they changed.
- The root cause, the fix, and how I verified it (which tests passed).

Why it matters to other people:

- Traceability. Anyone can see what was wrong and what was replaced, by serial number.
- Shift handoffs. The next technician can pick up where I left off without repeating my work.
- Patterns. Engineering and quality can spot the same failure across units and batches, like the same slot or the same part lot.
- Faster repairs. The next time it happens there is a known fix.
- RMA and warranty claims with the supplier need this data.
- Quality and audit requirements, and the customer's trust that the unit was fixed properly.

15

A server's fans spin and its health LED is green, but you never get a login prompt. Describe your troubleshooting stage by stage, and where you would look at each stage.

Fans and a green LED only prove the server has power. Booting is a chain, so I want to see how far it gets. I would open the console through the BMC (virtual console or SOL), watch a full boot, note exactly where it stops, and check the BMC event log for CPU, memory, PCIe or power errors.

1. POST: stuck on a POST code or beeping? Memory training on a large server can take several minutes, so I would give it time first. If it is really stuck, look up the code and suspect memory, a CPU or a PCIe card, and reseal. If needed, strip it to a minimum configuration (one CPU, one DIMM, no cards) and add parts back one at a time.
2. UEFI/BIOS: "No bootable device" or a PXE loop? Check the boot drive shows in the BIOS, the boot order, UEFI versus Legacy mode, Secure Boot, the cables and the RAID controller.
3. Bootloader: a `grub rescue>` prompt or "file not found" points to the boot partition or the drive.
4. Kernel: a kernel panic or "Unable to mount root fs" points to a storage driver, the root filesystem or the drive. Read the exact message.
5. System startup: emergency mode usually means a bad `/etc/fstab` entry or a failing disk. Log in on the console and read `journalctl -xb`.
6. If the console shows a login prompt but I cannot reach it over the network, the server has booted and it is a network problem instead.

At every stage: ask what changed, check firmware, and document where it stopped and what fixed it.

16

A network card shows in `lspci` with its driver loaded, and its interface appears in `ip a`, but the server cannot reach anything through it. Describe your process in order.

1. Link: does `ip a` show LOWER_UP or NO-CARRIER? Check `ethtool ens1f0` for "Link detected" and the speed, and the LEDs on both the NIC and the switch. No link means the cable, the transceiver (right type and speed), a dirty fiber, or a wrong or disabled switch port. Try a known-good cable or port.
2. Interface state: if it is DOWN, bring it up with `ip link set ens1f0 up` and check NetworkManager.
3. IP settings: is there an address? None, or 169.254, means DHCP failed. Check the subnet mask, look for a duplicate IP, and make sure the config is not pointing at an old interface name.
4. Local network: is the switch port on the right VLAN? Is ARP working (`ip neigh` shows the gateway's MAC, or FAILED)? Does the MTU match the switch? If small pings work but large transfers hang, it is an MTU mismatch.
5. Routing: `ip route` for the default gateway, ping the gateway, then ping beyond it and `traceroute`.
6. Firewall rules on the server.
7. Driver and firmware versions with `ethtool -i`, and error counters with `ethtool -S`. Compare with a working port or a working server.
8. Change one thing at a time, confirm it can reach the gateway and beyond, and document.

17

During a long stress test, an NVMe drive vanishes from `lsblk`. After a reboot it is back and looks healthy. What evidence would you look for, and what could be causing it?

A reboot resets the link, so the drive comes back, but whatever caused the drop is still there.

Evidence:

1. Kernel messages from when it dropped. If it has not been rebooted yet I save `dmesg > nvme_drop.txt` first; after a reboot I use `journalctl -b -1 -k | grep -i nvme`. I look for "I/O timeout, aborting", "controller is down; will reset" and "Removing after probe failure", plus PCIe AER errors.
2. The BMC event log for drive, PCIe or temperature events at the same time.
3. Drive health: `nvme smart -log /dev/nvme0` for temperature, critical warnings, media errors and unsafe shutdowns.
4. The link: `sudo lspci -vv` for `LnkSta` versus `LnkCap` on the drive, and AER errors on the port above it.
5. Firmware versions for the drive, the backplane and the BIOS.

Possible causes: poor seating or a worn backplane connector, a bad or misrouted cable, the drive overheating under load, a power-saving feature that does not wake up properly, a firmware bug, a PCIe signal problem, or the drive itself failing.

Then isolate: move the drive to another bay. If the problem follows the drive, it is the drive. If it stays with the bay, it is the bay, backplane or cable. Reproduce it under load (for example with `fio`) while watching `dmesg -w` and the temperature, and document everything.

18

An 8-GPU server shows only seven GPUs. In `lspci -tv`, the PCIe port that should lead to GPU 6 is present but has nothing under it. List what could cause this, and explain how you would narrow it down.

An empty port means the PCIe port is there but the device under it never linked. The path up to that port is working (otherwise the whole branch would be missing), so the problem is at the GPU end.

Possible causes:

- The GPU is not fully seated.
- Its auxiliary power cable is missing, loose or in the wrong connector.
- A bad slot or connector.
- A failed retimer on that path (on the riser or GPU baseboard).
- The slot disabled in the BIOS.
- The GPU itself is dead.

Narrowing it down:

1. Use the topology diagram to confirm which slot GPU 6 is in, and check the BMC inventory and SEL for power or PCIe events on that GPU.

2. Check dmesg for PCIe errors on that port, and the BIOS slot settings.
3. Powered off with ESD protection: reseal GPU 6 and check its power cable goes to the right connector. Check the PSU status.
4. Swap test: move GPU 6 to a working slot and put a known-good GPU in slot 6. If the problem follows the GPU, the GPU is bad. If it stays with slot 6, it is the slot, the power cable or the retimer.
5. Verify all eight GPUs show in lspci and nvidia-smi, run a full GPU test, and document.

19

You move a suspect GPU from Server A into known-good Server B, and Server B now fails the same way. What does this prove, what does it not prove, and what do you do next with both servers and the GPU?

It proves the GPU is very likely bad, because the failure moved with it into a system that was working.

It does not prove Server A is healthy. A bad slot or a power problem in Server A could have damaged the GPU in the first place, and it could damage the next good GPU that goes in.

Next steps:

1. Tag the GPU as bad and quarantine it so nobody reuses it.
2. Put Server B back to its original state with its own GPU and retest it, to make sure it still passes.
3. Inspect Server A's slot (pins, damage, burn marks) and its power cable, then test Server A with a known-good GPU.
4. Document the serial numbers of both servers and the GPU, the slots, and all the test results.
5. Start the RMA for the GPU, with failure analysis if needed.

20

Three servers from the same shift fail test because a cable was plugged into the wrong connector during assembly. After fixing them, what do you do, and how do you keep it from happening again?

1. Verify each of the three servers fully passes test after the fix, and that the wrong connection did not damage anything.
2. Document it: the unit serials, which cable, the wrong port and the correct port, the test results, and photos if allowed.
3. Notify my lead, quality and the assembly team.
4. Contain it. Three units from one shift means there could be more. Check other units from the same line, shift or batch, including ones that already passed and ones waiting to ship.
5. Prevent it:
 - Update the work instruction with clear pictures of the correct connection.
 - Add labels or color coding to the cable and the connector.
 - Use mistake-proofing, like keyed connectors or cable lengths that only reach the right port.
 - Add an inspection step that checks this cable before test.

6. Track it through the site's corrective action process (CAPA) so it gets followed up.
The focus is on fixing the process, not blaming the person who plugged it in.

Set B

Questions 21 to 40

21

Explain how a Linux server turns the name db01 into an IP address. Include which files are involved, the order they are checked, and what DNS records and caching are.

1. Linux checks `/etc/hosts` first. It is a local file that maps names to IP addresses. If db01 is in there, that answer is used and nothing else is asked. (The order is set in `/etc/nsswitch.conf`, normally files first, then DNS.)
2. If it is not in the hosts file, Linux asks the DNS server listed in `/etc/resolv.conf` (the "nameserver" line). If I typed a short name, the search domain from `resolv.conf` is added, for example db01.lab.local.
3. The DNS server answers from its own records if it is in charge of that domain. If not, it asks other DNS servers and passes the answer back.

DNS records are the entries stored on the DNS server. An A record maps a name to an IPv4 address, for example db01 to 10.4.2.15. (There are others, like AAAA for IPv6 and CNAME for an alias.)

Caching means the answer is saved for a while (its TTL) so the next lookup is faster and does not have to go all the way out again. It also means that when a record changes, old answers can stick around until the cache expires.

To test it I would use `nslookup db01` or `dig db01` (these ask DNS directly), and `getent hosts db01` to see what the system really uses, including `/etc/hosts`.

22

Your service guide includes a drawing of how every PCIe slot is wired, and Linux can print the PCIe tree it actually found. Explain what each one shows, which command prints the tree, and how comparing the two helps you find a fault.

The drawing is the topology diagram, and it is the expected picture. It shows which CPU and root port each slot connects to, which PCIe switches, risers, retimers and cables sit in between, the expected link width, how slots map to bus numbers, and which devices share a path.

The tree is the actual picture, what Linux really found. `lspci -tv` prints it, showing the root ports, the switches and the devices under them, with bus numbers like [17-1a].

Comparing them tells me where to look based on what is missing:

- A port is there but nothing is under it: that one device did not link. Suspect the device, its power cable, its seating, the slot or a retimer.
- A whole branch is missing: something upstream failed. Suspect the riser, cable, switch, retimer, the CPU root port, or a slot disabled in the BIOS.
- Everything under one switch is missing: suspect the switch or its upstream link, not each device.

So instead of swapping parts one by one, I can go straight to the part the missing devices have in common.

23

What is a BMC, and why does it keep working when the server's operating system has crashed? List at least six things a technician can do with it.

The BMC (Baseboard Management Controller) is a small, separate computer on the motherboard with its own processor, firmware and network port. Dell calls it iDRAC, HPE calls it iLO, Lenovo calls it XCC, and Supermicro and others call it IPMI or just the BMC.

It keeps working when the OS has crashed because it does not depend on the OS at all. It runs on standby power, which is there as long as the server is plugged in, and it has its own network connection. That is called out-of-band management. It even works with the server powered off or with no OS installed.

Things a technician can do with it (through its web page, `ipmitool` or Redfish):

1. Read the System Event Log (SEL): memory, PCIe, power, fan and temperature events with timestamps (`ipmitool sel elist`).
2. Read sensors: temperatures, voltages, fan speeds, PSU status and power draw (`ipmitool sensor`).
3. See the hardware inventory and health, serial numbers (`ipmitool fru print`) and firmware versions.
4. Use the virtual console (KVM) to see the screen.
5. Use Serial Over LAN for a text console (`ipmitool sol activate`).
6. Control power: on, off, reset or power cycle (`ipmitool power cycle`).
7. Send an NMI to force a crash dump.
8. Change the boot order and mount an ISO as virtual media.

24

What is a PCIe retimer, how is it different from a redriver, why are retimers so common in Gen 5 servers, and how might a failed retimer show up during testing?

A retimer is a chip in the PCIe path that fully recovers the data and re-sends a clean copy of the signal. That resets the loss budget, so the link can reach much farther across traces, connectors, risers and cables. It takes part in link training, it is normally invisible to the OS (it does not show in `lspci`), and it sits on risers, backplanes, motherboards and GPU baseboards.

A redriver is just an amplifier. It boosts the signal, but it boosts the noise along with it, and it does not take part in link training.

They are common in Gen 5 because at 32 GT/s the signal loses strength much faster over the same distance. Long paths to front drive bays, risers and GPU trays cannot work at Gen 5 without help, and a redriver is not good enough at those speeds.

How a failed retimer can show up:

- Every device behind it disappears, so a whole branch is missing in `lspci -tv`.
- Links come up downtrained (LnkSta lower than LnkCap).
- A steady stream of corrected PCIe (AER) errors in `dmesg`.
- Devices dropping out under load.

Since it does not appear in `lspci`, I would check the BMC (some report retimer health) and the retimer firmware version.

25 Explain what `LnkCap` and `LnkSta` mean in `lspci -vv` output. What does it mean when they do not match, and what could cause it?

`LnkCap` is the link capability: the fastest speed and widest width the device can do, for example Speed 16GT/s (Gen 4), Width x16.

`LnkSta` is the link status: what the link actually trained to when the system powered on.

Every PCIe link negotiates (trains) its speed and width at power on. If `LnkSta` matches `LnkCap`, the device got its full link. If `LnkSta` is lower, the link is downtrained. It still works, just slower. For example, `LnkCap` x16 and `LnkSta` x4 means a quarter of the bandwidth.

Possible causes:

- The slot or path cannot do more, like a Gen 5 card in a Gen 4 slot, or an x16 slot that is only wired x8. That one is expected, not a fault.
- Poor seating or a dirty connector.
- A bad riser, cable or retimer.
- A BIOS setting forcing a lower speed, or the wrong bifurcation.
- Signal problems, which usually also show up as AER errors in `dmesg`.

I would check the topology diagram first to see what the slot should give, then reseal and clean, swap the riser or cable one at a time, and check `LnkSta` again after each change.

26 A server crashed overnight and rebooted itself. Which one shows you what happened before the crash, `dmesg` or `journalctl -b -1`? Explain what each one shows, and what you should save before you reboot a failing server.

`journalctl -b -1` shows what happened before the crash.

`dmesg` shows the kernel messages from the current boot only. It reads a buffer in memory that is cleared on every reboot, so once the server rebooted itself, the messages from before the crash are gone from `dmesg`.

`journalctl` reads the systemd journal. `-b -1` means the previous boot, so it shows the messages leading up to the crash (add `-k` for kernel messages only). This only works if the journal is kept on disk. If it says there is no previous boot, that evidence is lost.

At the end of the previous boot I would look for hardware errors or a kernel panic. If the log just stops with no shutdown messages, that points to a sudden power loss or a hardware reset, so I would check the BMC event log for the same time.

Before rebooting a failing server I would save:

- The kernel messages: `dmesg -T > before_reboot.txt`

- Any journal output I need.
- The BMC event log and sensor readings.
- A screenshot of the console if it is showing an error.

Then copy those files off the server before the reboot.

27 Describe the four layers you check to decide whether a device is really working in Linux, and name the command or check you would use at each layer. Give an example where a device passes one layer and fails the next.

1. Physical: is it installed, seated and powered, and does the BIOS or BMC see it? Check by looking at it, its LEDs, and the BIOS or BMC inventory.
2. Bus: does the OS see it on the bus? `lspci` for PCIe devices, `lsblk` or `nvme list` for drives.
3. Driver: is a driver attached? `lspci -k` shows "Kernel driver in use", and `dmesg` shows driver errors.
4. Function: is it configured and actually working? `ip a` for a NIC, `nvidia-smi` for a GPU, or a functional test.

Example: a NIC shows up in `lspci`, so it passes the bus layer, but `lspci -k` has no "Kernel driver in use" line, so it fails at the driver layer. The hardware is fine and the problem is the driver, so the fix is to install or load the driver, not to replace the card.

Another one: a GPU shows in `lspci` but not in `nvidia-smi`. That is a driver or firmware problem, not hardware.

28 Explain the swap with known-good technique. What does it mean if the problem follows the part, what does it mean if it stays with the slot, and what are the risks of the technique?

Swap with known-good means moving the suspect part into a slot or system that is known to work, or putting a known-good part into the suspect slot, and retesting. Only that one thing changes.

- If the problem follows the part, the part is bad.
- If the problem stays with the slot (a good part now fails there too), the problem is the slot, cable, riser or board, not the part.

Risks:

- A bad slot or power problem can damage the good part you put in, and then you have two bad parts. I would inspect the slot first.
- A damaged part could damage the good system you move it into.
- It does not prove everything. If the part fails in the good system, that still does not prove the original system is fine, since the original slot may be what damaged the part.
- Handling has its own risks: ESD, bent pins, and the fact that just reseating a part can temporarily "fix" a seating problem and hide the real cause.

- Parts can get mixed up, so I would tag them and record the serial numbers.

29

You are given a failure you have never seen before, with an error message you do not recognize. Describe your approach, including when and how you would escalate.

1. Stay calm and do not guess or start swapping parts.
2. Write down the exact symptoms and the exact error message, word for word, and when it happens.
3. Gather logs: `dmesg`, `journalctl`, the BMC event log and sensors, plus serial numbers and firmware versions. Save them before any reboot.
4. Read the service guide and the documentation for the system and the part.
5. Search the exact error message (internal knowledge base, vendor documentation).
6. Break the problem into layers (physical, bus, driver, function) to find where it fails.
7. Test one theory at a time, simplest first, changing one thing at a time and recording results.
8. Escalate when I am stuck, when the next step is beyond what I am allowed to do, or when there is a risk of damage or a safety issue. I would go to my lead or engineering with my data: the symptom, the logs, what I tested and what happened, so they do not have to start over.
9. Once it is solved, document what I learned so the next person has the answer.

30

A server's operating system is hung, and you are asked to hard reset it right away. What would you capture first, where would you get it, and why is it worth the time?

Before the reset I would take a few minutes to collect evidence through the BMC, because a reset wipes what is on the screen and in memory.

1. Look at the console with the BMC virtual console or SOL. Is there a kernel panic or error on the screen? Take a screenshot.
2. Check the BMC event log (`ipmitool sel elist`) for memory, PCIe, power or temperature events around the time it hung.
3. Check the sensors and health in the BMC: temperatures, fans, PSU status and power.
4. Save it all: export the event log and sensor readings and keep the screenshot.
5. If `kdump` is set up, send an NMI from the BMC (`ipmitool chassis power diag`) instead of a plain reset, so the kernel writes a crash dump that can be analyzed.
6. Then power cycle, and after it boots read `journalctl -b -1` for the last messages from the hung boot.

It is worth the time because this evidence only exists until the reset. With it we have a chance of finding the real cause. Without it we just reset and wait for it to hang again.

31

An NVMe drive shows LnkSta: Speed 8GT/s, Width x2 while its LnkCap shows Speed 16GT/s, Width x4. The drive works but is slow. How would you investigate?

The drive can run Gen 4 x4 but trained at Gen 3 x2, so it is getting about a quarter of its bandwidth (about 2 GB/s instead of about 8 GB/s).

1. Check the topology diagram or service guide: is that bay supposed to give Gen 4 x4? If the slot is only Gen 3 or wired x2, this is expected.
2. Check the BIOS: is the link speed forced to Gen 3, or is the bifurcation setting wrong for that slot or adapter?
3. Check `dmesg` for corrected PCIe (AER) errors on the drive's address and on the port above it. A steady stream points to a signal problem.
4. Check firmware versions for the drive, the backplane and the BIOS.
5. Physical, powered off with ESD protection: reseal the drive, inspect and clean the connector, and check the backplane connector, the cable (MCIO or SlimSAS) and any riser or retimer on that path.
6. Isolate: move the drive to another bay. If it trains at full speed there, the problem is the original bay, backplane or cable. If it is still downtrained, it is the drive.
7. Change one thing at a time and recheck LnkSta after each change. Confirm the speed with a performance test (for example `fio`) and document it.

32

Compare these two cases: (a) ping 10.4.2.15 works but ping db01 fails with "Name or service not known"; (b) ping db01 resolves to 10.4.2.15 but gets no replies. What does each result point to, and what would you check in each case?

(a) The IP works, so the network is fine. The problem is name resolution. I would check:

- `/etc/resolv.conf`: is the right DNS server listed?
- Can I reach that DNS server at all?
- `nslookup db01` or `dig db01` to ask DNS directly.
- `/etc/hosts` for a missing or wrong entry.
- Whether I need the full name (FQDN), for example `db01.lab.local`, or the right search domain.

(b) The name resolved, so DNS is working. The problem is reaching the address, or db01 itself. I would check:

- Is 10.4.2.15 still the right address for db01? An old DNS record or hosts entry could point to the wrong place.
- My own side: `ip a`, `ip route`, and ping the gateway.
- `traceroute 10.4.2.15` to see where the traffic stops.
- Is db01 actually up? Check its console or BMC.
- Is a firewall on db01 or in between blocking ping?

33

In a dual-socket server, every PCIe device connected to CPU 2 is missing, while everything on CPU 1 works. What could cause this, and how would you narrow it down?

In a dual-socket server each PCIe slot belongs to one CPU, and that CPU's root complex provides the lanes. If everything on CPU 2 is missing, the common factor is CPU 2 or something it depends on, not the individual devices.

Possible causes:

- CPU 2 not seated properly, or bent pins in its socket.
- CPU 2 has failed or is not being initialized.
- A BIOS setting disabling the slots or root ports on CPU 2, or the wrong bifurcation.
- A riser or cable that carries all of CPU 2's lanes.
- A power problem on the CPU 2 side of the board.

Narrowing it down:

1. `lscpu`: does it show both sockets and all the cores? `free -h` and `dmidecode -t memory`: is CPU 2's memory there? If CPU 2's memory is missing too, it is almost certainly the CPU or its socket.
2. Check the BMC event log and the BIOS for CPU, memory or PCIe errors on CPU 2.
3. Check the BIOS slot and bifurcation settings.
4. Compare `lspci -tv` with the topology diagram to confirm every missing device sits behind CPU 2's root ports.
5. Powered off with ESD protection: reseal CPU 2, inspect the socket for bent pins, and check the risers and cables.
6. Swap CPU 2 with a known-good CPU and see whether the problem follows the CPU. Verify everything is back and document.

34

A server should have 1 TB of memory, but `free -h` shows about 960 GB. How would you find what is missing and why?

`free -h` always shows a little less than what is installed, because some memory is reserved by the firmware and kernel. But 1 TB is 1024 GB, and 1024 minus 960 is 64 GB, which is exactly one 64 GB DIMM. So I suspect one DIMM is missing or disabled.

1. `sudo dmidecode -t memory`: go through every slot and look for "No Module Installed" in a slot that should be filled, or a DIMM with the wrong size. It also gives each DIMM's serial number.
2. Count the DIMMs and their sizes against the build sheet.
3. Check the BIOS memory screen for a DIMM that is disabled or mapped out because of errors.
4. Check the BMC event log and `dmesg | grep -i edac` (or `ras-mc-ctl --summary`) for memory errors on that slot.
5. Check the population rules in the service guide, since a DIMM in the wrong slot can be disabled.

6. Powered off with ESD protection: reseal the DIMM with both latches closed, and make sure it is the right part (same type, size and speed as the others).
7. If it still does not show, swap it with a known-good DIMM to see whether the DIMM or the slot is bad. Then confirm the full amount shows, run a memory test, and document.

35

Here is part of the `ip a` output from a server that cannot reach anything on the network. What does it tell you, what is most likely wrong, and what would you check next?

```
3: ens2f1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc mq state UP  
    link/ether b8:ce:f6:4a:10:22 brd ff:ff:ff:ff:ff:ff  
    inet 169.254.88.4/16 brd 169.254.255.255 scope link ens2f1
```

What it tells me:

- The interface `ens2f1` is UP and has a physical link (`LOWER_UP`), so the NIC, cable and switch port are at least passing a link.
- The MTU is 9000 (jumbo frames).
- The MAC address is `b8:ce:f6:4a:10:22`.
- The address is `169.254.88.4/16` with "scope link". That is a self-assigned, link-local address. It means the server asked DHCP for an address and got no answer, so it gave itself one. There is no "dynamic" either, so there is no DHCP lease.

Most likely problem: DHCP failed. With a `169.254` address the server cannot reach anything beyond its own link.

What I would check next:

1. Is the switch port on the right VLAN? If it is on a network with no DHCP server, nobody will answer.
2. Is the DHCP server up, and does it have a free address or a reservation for this MAC?
3. If the DHCP server is on another subnet, is the DHCP relay set up on the router?
4. Is the cable in the right NIC port? On a dual-port card the other port may be the one that is configured.
5. Does the MTU of 9000 match the switch?
6. Then renew the address (for example `sudo dhclient ens2f1` or restart the connection in NetworkManager) and confirm it gets a real address, a gateway in `ip route` and DNS servers. Document the fix.

36

One port of a dual-port network card works. The other has a link and a valid IP address on the same subnet as its gateway, but small pings succeed while large file transfers hang. What do you suspect, and how would you confirm it?

I would suspect an MTU mismatch. For example, the port is set to 9000 (jumbo frames) but the switch port or something in the path only allows 1500. Small packets fit, so pings work, but full-size packets get dropped, so big transfers hang.

To confirm:

1. Compare the MTU of both ports with `ip a`. The working port probably matches the switch and the broken one does not.
2. Ping with large packets that are not allowed to be split up: `ping -M do -s 8972 <gateway>` (8972 bytes plus 28 bytes of headers is 9000). If that fails but `ping -M do -s 1472 <gateway>` works, the path only supports 1500.
3. Check the MTU setting on the switch port (with the network team).
4. Check `ethtool -S` for errors or drops on that port.

Then set the MTU to match on both sides, change only that, and confirm a large file transfer works. If the MTU already matches, I would look at the VLAN and error counters next. Document the result.

37

In a 4-GPU server, the GPU in slot 2 disappears from `nvidia-smi` and `lspci` after 30 to 60 minutes of a stress test, and comes back after a cold power cycle. Before it vanished, `dmesg` showed "GPU has fallen off the bus". Describe the data you would collect, the hypotheses you would test in order, and how you would prove the root cause.

"Fallen off the bus" (Xid 79) means the GPU stopped responding on PCIe. It happening only after 30 to 60 minutes under load points to heat, power or a weak signal.

Data to collect:

- `dmesg / journalctl -k` with timestamps: the Xid 79 message and any PCIe AER errors before it.
- The BMC event log and sensors at that time: GPU and inlet temperatures, fan speeds, PSU status and power draw.
- `nvidia-smi -q` for that GPU's temperature, power, serial number, VBIOS version and ECC errors, and `watch -n 1 nvidia-smi` during the test.
- `LnkSta` versus `LnkCap` for the slot 2 GPU.
- Firmware and driver versions, serials, and whether other servers have the same problem in slot 2.

Hypotheses, simplest first:

1. Overheating: blocked airflow, a missing baffle or blank, a failed fan, or slot 2's position. Watch the temperature climb before it drops.
2. Power: a loose or bad auxiliary power cable, or a PSU problem.

3. Seating or signal: the GPU, riser, cable or retimer, made worse as things heat up.
4. A firmware or driver bug.
5. The GPU itself.

Proving it:

- Reproduce it on purpose with the stress test while watching the temperature and `dmesg -w`.
- Change one thing at a time (reseat and recable, fix airflow) and rerun the full-length stress test.
- Swap the slot 2 GPU with one from a working slot. If the failure follows the GPU, it is the GPU. If it stays in slot 2 with a good GPU, it is the slot, cable, riser, cooling or power.
- The cause is proven when the failure moves (or stays) as expected and the fix passes repeated full stress runs. Document everything.

38

A server booted fine yesterday. Overnight, someone replaced the PCIe adapter card that holds the boot NVMe drive, and now the console shows the BIOS logo and then “No bootable device”. Describe your approach.

It worked yesterday and the only change was the adapter card, so I start there.

1. Find out exactly what was done: which card came out, which went in, whether it is the same model, and whether the boot drive was moved over correctly. Check the BMC event log too.
2. Go into the BIOS. Does it see the NVMe drive at all?
3. If it does not: power off with ESD protection and check the drive is seated on the adapter, the adapter is fully seated in the right slot, and the riser is seated. Check the slot is enabled in the BIOS and the bifurcation is right for that adapter (a multi-drive card needs the slot split into x4 links).
4. If the BIOS does see the drive: check the boot order. A new adapter or slot can lose the boot entry, so the drive may need to be selected again as the boot device.
5. Check the boot mode (UEFI versus Legacy) and Secure Boot, in case they were changed.
6. Check the firmware on the new adapter, or if it is a RAID adapter, that the volume is configured.
7. If it still will not boot, put the original card back to confirm the new card is the problem.

Once it boots, verify the system fully, and document what was wrong and what fixed it.

39

After a firmware update, a Linux server stops responding to ping and SSH, but its BMC is reachable. How do you work out whether the problem is the hardware, the operating system, or the network?

Since the BMC works, I would use it to look at the server without depending on the OS or its network.

1. Open the console (virtual console or SOL). Is it still updating, stuck in POST, at a BIOS screen, at a kernel panic, or sitting at a login prompt? If an update is still running, I do not power it off.
2. Check the power status, the event log and the sensors in the BMC. If it is stuck in POST or the event log shows hardware errors, it is a hardware or firmware problem.

3. If it booted to the OS, log in through SOL and check from the inside:
 - `lspci` and `lspci -k`: is the NIC there, and is its driver loaded? A NIC firmware update can need a newer driver.
 - `ip a`: is the interface up with an IP? Did the interface name change, leaving the config pointing at the old name?
 - `ip route`, then ping the gateway.
 - `systemctl status sshd`.

If the OS cannot see the NIC or the driver fails, it is the OS or driver (or the NIC). If the NIC looks fine and has an IP but cannot reach the gateway, it is the network side: the switch port, VLAN or cable.
4. Compare the firmware versions with the approved list and the update notes, and check the documented order was followed (for example BMC before BIOS).
5. Fix one thing at a time, retest ping and SSH, and document the firmware versions before and after.

40

A server randomly reboots once or twice a day, with nothing obvious in the operating system logs. Describe what you would check and how you would catch the cause.

Nothing in the OS logs is actually a clue. If the journal just stops with no shutdown messages, the OS did not choose to reboot. Something underneath reset it: power, hardware, or a watchdog.

What I would check:

1. `journalctl --list-boots` to get the reboot times, and `journalctl -b -1` to read the last messages before each one.
2. The BMC event log at those exact times: power or PSU events, thermal events, uncorrectable memory errors, CPU errors, or a watchdog reset.
3. Power: are both PSUs healthy and redundant? Check the power cords, the outlets or PDU, and whether both PSUs are on the same circuit.
4. Temperatures and fan speeds from the BMC sensors.
5. Whether a watchdog is enabled and resetting the server.
6. BIOS and BMC firmware versions for known issues.
7. Whether the reboots line up with load or a time of day, like a scheduled job.

How I would catch it:

- Make sure the journal is kept on disk and set up `kdump`, so a crash leaves evidence.
- Keep an SOL session logging to a file, so I can see the console at the moment it resets.
- Watch the sensors and power over time, and try to trigger it with a stress test.
- Change one thing at a time (for example swap one PSU) and give it enough days to trust the result. Compare with a golden unit, and document everything.